

PREM RAICHURA

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[Github](#) | [Linkedin](#) | [Portfolio](#)

PROFILE SUMMARY

A Full-Stack & AI Developer with a strong focus on building intelligent, scalable, and user-centric solutions. I enjoy transforming complex ideas into clean, functional systems — from robust backend architectures to intuitive frontend interfaces. My experience spans across AI/ML-driven applications, deep learning research, and web development using technologies such as FastAPI, React, PHP, Python, and MySQL.

EDUCATION

L.D.R.P. Institute of Technology and Research.

Bachelor of Engineering in Computer Engineering.

2024-2027

Government Polytechnic Porbandar.

Diploma in Computer Engineering. - CGPA: 7.09

2020-2023

SKILLS

- **Backend:** Php, Laravel, Python, Django, Flask, RESTful, Fast APIs, Java, Android, ReactNative, SQL, C, C++
- **Frontend:** React.js, Bootstrap, Tailwind CSS, JavaScript, HTML/CSS
- **ML/DL Frameworks:** TensorFlow, PyTorch, Pandas, NumPy, OpenCV, Keras
- **Version Control:** Git, GitHub, Docker
- **Database:** MySQL, Firebase, Supabase
- **Others:** Postman, Vscode

COURSEWORK

Core Concepts: Data Structures and Algorithms, Operating Systems, OOP, DBMS, Computer Networks

EXPERIENCE

INTERNSHIP AT INSTITUTE OF PLASMA RESEARCH (IPR)

29 DEC 2026 - Ongoing

> Role: Devop's & Backend Developer

◇ Compared & reviewed multiple ERP systems for the implementation based on complexity, cost efficiency & scalability.

◇ Deployed, Configured & customized ERPNext system as per the IPR's requirement. Implemented Modules: Attendance, Multiple Dashboards & Auto Mailings

PUBLICATION

TRAFFICEYE: INTELLIGENT TRAFFIC OPTIMIZATION USING DEEP LEARNING APPROACH



> Published in IEEE International Conference on Artificial Intelligence and Machine Vision (AIMV 2025)

◇ Built and evaluated an AI-driven traffic optimization system using YOLOv5s on the IITM-HeTra dataset for real-time vehicle detection and analysis.

◇ Contributed to model development, experimentation, and results validation, leading to successful publication in IEEE AIMV 2025.

PROJECTS

TRAFFIC DENSITY & DETECTION WEB APP



› *React + vite, FastAPI, YOLOv5s, AI/ML*

- ◇ Developed a real-time traffic monitoring web app using FastAPI, React + vite, and YOLOv5s trained on the IITM-HeTra dataset.
- ◇ Achieved 71 % accuracy and 1.4 ms inference through an AI/ML-optimized pipeline for real-time traffic analysis.

WEALTHEQUITY



› *React + vite, Tailwind CSS, Email.js*

- ◇ Developed the official WealthEquity website using React, Vite, and Tailwind CSS with a responsive, animated UI for an engaging user experience.
- ◇ Integrated Email.js for automated inquiries, boosting engagement and contributing to a 5+ client increase through improved digital presence.

PROMPT-INJECTION-PREVENTION



› *Python, pytorch, ViT*

- ◇ Developed a hybrid AI safety system using rule-based filters and a ViT classifier to detect unsafe prompts, achieving 98.32% accuracy.
- ◇ Trained and optimized a Vision Transformer (ViT) hybrid model on labeled prompt datasets to identify harmful patterns and enhance LLM safety with high reliability.

ACHIVEMENTS

RESEARCH PROJECT GRANT



› *Titled "SecurePark — The Real Edge AI, Vision and IoT-based campus security and Parking ecosystem for KSV."*

- ◇ Awarded a project grant of ₹69,017/- by M. M. Patel Students Research Project Cell, KSV

RESEARCH PAPER GRANT



› *Titled "TrafficEye: Intelligent Traffic Optimization Using Deep Learning Approach."*

- ◇ Awarded a conference grant of ₹2,500/- by M. M. Patel Students Research Project Cell, KSV

INNOVATION - DAU HACKATHON 2026



› *Finalist*

- ◇ Developed an Early Disease Detection System From Medical Imaging

ANRF'S AISEHACK 2026

› *Finalist*

- ◇ Finetuned a Prithvi-100m model for the flood detection on the dataset given by IBM. Achieving a 0.703 score satisfies the minimum criteria.

DAU - TIC TAE TOE 2024



› *Finalist*

- ◇ Developed an E-learning Platform : EDUMATE

CERTIFICATIONS

- ◇ NPTEL - PYTHON FOR DATA SCIENCE (ELITE).
- ◇ BASIC & ADVANCE PYTHON.

